

PS-BEFORE-MID — Weeks 1–8 · 17 Aug – 11 Oct 2026 ·

MIDTERM (W8)

Scope: Weeks 1–8 (17 Aug – 11 Oct 2026) — descriptive statistics, central tendency & dispersion, probability, conditional probability & Bayes, random variables, probability distributions, ending at the Week 8 midterm.

Siblings: [PS-NAV.md](#) (master map) · [PS-AFTER-MID.md](#) (Weeks 9–15). Week-colored headers below match the NAV schedule. **Exam split:** 60% numerical + 40% theory. Formula mastery + problem-solving speed is the skill.

Exam order: numerals first, theory second. Z-table fluency is the gate.

At a glance (W1–W8)

W	Dates	Variant	File	Topics	Chapter
W1	17–23 Aug	P2	BEFORE	Descriptive statistics	01
W2	24–30 Aug	P2	BEFORE	Central tendency & dispersion	02
W3	31 Aug–06 Sep	P2	BEFORE	Probability theory	03
W4	07–13 Sep	P2	BEFORE	Conditional probability & Bayes	03
W5	14–20 Sep	P2	BEFORE	Random variables & random processes	04
W6	21–27 Sep	P2	BEFORE	Probability distributions	05
W7	28 Sep–04 Oct	P1	BEFORE	Midterm revision	01–05
W8	05–11 Oct	MIDTERM	BEFORE	Exam week — no new material	—

Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week.

W1 — Descriptive Statistics · 17–23 Aug · P2 · 4 hrs

Banner: Rotation PS (Mon+Tue) · Tier P2 · Time budget 4 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 1	lines 17–32
Fear-Killer Pack	Fear-Killer-Packs.md → Week 1	lines 11–16
Week manifest	weeks/PS-W1.md	full file
Chapter breakdown	03-Chapter-Breakdowns/01-Introduction-to-Statistics.md	embedded below

Definitions (verbatim)

Term	Definition
Statistics	The science of collecting, organizing, presenting, analyzing and interpreting data to make decisions.
Variable	A characteristic that varies from one observation to another (e.g. height, score).
Frequency distribution	A table that groups data into classes and records the number (frequency) of observations falling in each class.
Class interval	The range of values grouped in one class of a frequency distribution.
Histogram	A graphical representation of a frequency distribution using adjacent rectangles whose areas are proportional to class frequencies.
Frequency polygon	A line graph connecting the class midpoints plotted at their frequencies.

Ogive / frequency curve	A cumulative frequency graph plotted against class boundaries.
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Formulas (verbatim)

Formula	Statement
Number of classes (Sturges)	$k = 1 + 3.322 \log_{10}(n)$
Class width	$(\text{max} - \text{min}) / k$
Class mark	$(\text{lower limit} + \text{upper limit}) / 2$
Relative frequency	f_i / N
Cumulative frequency	running sum of frequencies

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md · #2 histogram and frequency polygon · #3 box plot (min, Q1, median, Q3, max)
Numericals	Numerical-Book.md · #46 mean, median, mode (grouped/ungrouped) · #47 variance, standard deviation · #48 quartiles, IQR

Books · Chapters · Media

Resource	Where
Devore	Ch 1 (types of data, presentation)
StatQuest	Statistics Fundamentals — data types, graphs
Brandon Foltz	Statistics 101 — Descriptive Stats playlist

Fear-Killer Pack Week 1 (verbatim)

Resources: Brandon Foltz (Descriptive Stats) • Devore Ch.1

- Given the dataset: 12, 15, 18, 20, 22, 25, 27, 30, 35, 42, 50, 55, 60, 65, 70. Calculate mean, median, mode, Q1, Q3, IQR, variance, standard deviation, and coefficient of variation. Now add an outlier of 200 and recompute all measures. Which measures are robust to outliers and which are not?
- A frequency distribution of exam scores for 50 students is: 30-40 (3), 40-50 (7), 50-60 (12), 60-70 (15), 70-80 (8), 80-90 (5). Calculate the mean for grouped data using the midpoint method. Then compute the variance for grouped data. Compare to if you had the raw data — what information is lost?
- Two datasets have same mean = 50 but Dataset A has variance = 25 and Dataset B has variance = 400. Dataset A is symmetric while Dataset B is positively skewed. Draw the approximate distributions. Interpret what each tells you about the data spread and shape. How does skewness affect which measure of central tendency is most representative?

Narrative — Week 1 (verbatim)

Topics: Types of data (qualitative, quantitative, discrete, continuous); Variables; Data presentation (tables, bar charts, pie charts, histograms, frequency polygons, ogives)

Resources: Devore Ch.1; StatQuest — Statistics Fundamentals (YouTube)

Practice: - Classify 10 data types (nominal/ordinal/interval/ratio) - Construct frequency distribution from raw data: class intervals, frequency, cumulative frequency, relative frequency - Choose the correct graph type for a given data set

Deliverable: Frequency distribution table + histogram for a given dataset

Time budget: 4 hrs

Chapter 1 — Introduction to Statistics (verbatim)

Weight: ★★☆☆☆

Difficulty: ★☆☆☆☆

Importance: ★★★★★

Learning Objectives

Explain

- Object and classifications of statistics
- Types of data & variables

Construct

- Frequency distributions (types)
- Tables (tabulation)

Draw

- Simple, multiple and component bar diagrams
- Pie-chart
- Histogram
- Frequency polygon and frequency curves

Differentiate

- Qualitative vs quantitative data
- Discrete vs continuous variables
- Frequency distribution types

Must Memorize

Statistics: The science of collecting, organizing, presenting, analyzing and interpreting data to make decisions.

Variable: A characteristic that varies from one observation to another (e.g. height, score).

Frequency distribution: A table that groups data into classes and records the number (frequency) of observations falling in each class.

Class interval: The range of values grouped in one class of a frequency distribution.

Histogram: A graphical representation of a frequency distribution using adjacent rectangles whose areas are proportional to class frequencies.

Frequency polygon: A line graph connecting the class midpoints plotted at their frequencies.

Ogive / frequency curve: A cumulative frequency graph plotted against class boundaries.

Must Understand

- Data → Table → Graph pipeline; the same data can mislead if the wrong graph is chosen
- Class boundaries vs class limits vs class marks
- Cumulative vs relative vs simple frequency
- When a bar chart (categorical) vs histogram (quantitative) is correct

Must Practice

Raw: 12, 15, 9, 21, 18, 10, 8, 16, 14, 20

Class (10–12, 13–15, 16–18, 19–21) → frequency, cumulative freq, relative freq

Common Mistakes

1. **Class limits vs boundaries** — boundaries sit 0.5 unit outside limits for integer data.
2. **Bar chart vs histogram** — bar charts have gaps (categories); histograms are contiguous (continuous data).
3. **Choosing too few/too many classes** — Sturges' rule: $k = 1 + 3.322 \cdot \log_{10}(n)$.
4. **Reading the frequency polygon at the wrong point** — plot at class midpoints, not limits.

Typical Questions

- 1. Classify the given variables as qualitative/quantitative, discrete/continuous.
- 2. Construct a frequency distribution from raw data; draw the histogram and frequency polygon.
- 3. Present categorical data as a pie-chart and component bar diagram.
- 4. Compare histogram vs bar diagram with a suitable example.
- 5. What are the objects of statistics?

Examiner Expectations

Level	Performance
Pass	Can construct a frequency table
Good	Draws correct histogram/polygon, chooses graph type
Excellent	Handles open-ended and unequal classes correctly
Full marks	Error-free tabulation under time pressure

Formula Sheet

Number of classes (Sturges): $k = 1 + 3.322 \log_{10}(n)$
Class width: $(\text{max} - \text{min}) / k$
Class mark (midpoint): $(\text{lower limit} + \text{upper limit}) / 2$
Relative frequency: f_i / N
Cumulative frequency: running sum of frequencies

GPA Priority: ☐ Essential (foundation, low weight)

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 robust measures + Q2 grouped mean) · def/formula skim · same-problem drill #46
- **Same-problem drill target:** grouped-data mean/variance < 10 min; frequency distribution from raw data clean
- **Deliverable:** frequency distribution table + histogram for a given dataset
- **Trap:** bar chart (categorical) vs histogram (quantitative); class boundaries sit 0.5 outside limits

W2 — Measures of Central Tendency & Dispersion · 24–30 Aug · P2 · 4 hrs

Banner: Rotation PS (Mon+Tue) · Tier P2 · Time budget 4 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 2	lines 34–51
Fear-Killer Pack	Fear-Killer-Packs.md → Week 1	lines 11–16 (Q1, Q3)
Week manifest	weeks/PS-W2.md	full file
Chapter breakdown	03-Chapter-Breakdowns/02-Measures-of-Central-Tendency-and-Dispersion.md	embedded below

Definitions (verbatim)

Term	Definition
Mean	The sum of all observations divided by their number.
Median	The middle value when data is arranged in order; Q2.
Mode	The most frequently occurring value.
Quartile deviation	Half the interquartile range, $(Q3 - Q1) / 2$.
Standard deviation	The positive square root of the variance — the average deviation of observations from the mean.
Coefficient of variation	The ratio of standard deviation to mean, expressed as a percentage — $\sigma/\bar{x} \times 100$.
Skewness	The degree of asymmetry of a distribution about its mean.

Kurtosis	The peakedness or flatness of a distribution relative to the normal.
Moments	The r-th moment about mean: $\mu_r = \sum f(x-\bar{x})^r / \sum f$.

Formulas (verbatim)

Formula	Statement
Mean (raw)	$\bar{x} = \sum x / n$
Mean (grouped)	$\bar{x} = \sum fx / \sum f$
Median (grouped)	$L + (n/2 - CF) / f \times h$
Mode (grouped)	$L + (f_1 - f_0) / (2f_1 - f_0 - f_2) \times h$
Variance (pop)	$\sigma^2 = \sum (x-\mu)^2 / N = \sum x^2/N - \mu^2$
Variance (sample)	$s^2 = \sum (x-\bar{x})^2 / (n-1)$
Quartile deviation	$(Q3 - Q1) / 2$
C.V.	$(\sigma / \bar{x}) \times 100$
Skewness (β_1)	μ_3^2 / μ_2^3
Kurtosis (β_2)	μ_4 / μ_2^2

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md #3 box plot (min, Q1, median, Q3, max)
Numericals	Numerical-Book.md #47 variance/SD (pop vs sample) · #48 quartiles, IQR

Books · Chapters · Media

Resource	Where
Devore	Ch 1 (central tendency, dispersion)
StatQuest	Mean, Median, Mode

Fear-Killer Pack Week 1 (verbatim — Q1, Q3)

Resources: Brandon Foltz (Descriptive Stats) • Devore Ch.1

- Given the dataset: 12, 15, 18, 20, 22, 25, 27, 30, 35, 42, 50, 55, 60, 65, 70. Calculate mean, median, mode, Q1, Q3, IQR, variance, standard deviation, and coefficient of variation. Now add an outlier of 200 and recompute all measures. Which measures are robust to outliers and which are not?
- Two datasets have same mean = 50 but Dataset A has variance = 25 and Dataset B has variance = 400. Dataset A is symmetric while Dataset B is positively skewed. Draw the approximate distributions. Interpret what each tells you about the data spread and shape. How does skewness affect which measure of central tendency is most representative?

Narrative — Week 2 (verbatim)

Topics: Mean, median, mode; Quartiles, percentiles; Range, variance, standard deviation; Coefficient of variation; Skewness and kurtosis

Resources: Devore Ch.1; StatQuest — Mean, Median, Mode

Practice: - Compute mean, median, mode for grouped and ungrouped data - Compute variance and standard deviation (both population and sample formulas) - Interpret skewness: positive (tail right), negative (tail left), zero (symmetric) - Quartile deviation: $(Q3 - Q1) / 2$

Deliverable: Complete descriptive statistics for 2 datasets

Time budget: 4 hrs

Chapter 2 — Measures of Central Tendency & Dispersion (verbatim)

Difficulty: ★★☆☆☆

Importance: ★★★★★

Learning Objectives

Calculate

- Mean, median, mode (raw and grouped data)
- Quartiles, range, quartile deviation
- Mean deviation, standard deviation, variance
- Moments, coefficient of variation

Interpret

- Skewness & kurtosis
- Coefficient of variation

Differentiate

- When mean vs median vs mode is the best measure
- Population vs sample formulas

Must Memorize

Mean: The sum of all observations divided by their number.

Median: The middle value when data is arranged in order; Q2.

Mode: The most frequently occurring value.

Quartile deviation: Half the interquartile range, $(Q3 - Q1) / 2$.

Standard deviation: The positive square root of the variance — the average deviation of observations from the mean.

Coefficient of variation: The ratio of standard deviation to mean, expressed as a percentage — $\sigma/\bar{x} \times 100$.

Skewness: The degree of asymmetry of a distribution about its mean.

Kurtosis: The peakedness or flatness of a distribution relative to the normal.

Moments: The r-th moment about mean: $\mu_r = \Sigma f(x-\bar{x})^r / \Sigma f$.

Must Understand

- Which average is most representative depends on shape (skew pulls the mean)
- Grouped-data formulas use midpoints; information is lost vs raw data
- σ^2 (variance) is additive, σ is not
- C.V. allows comparing variability across different units/scales
- β_1 (skewness), β_2 (kurtosis) and their normal-distribution benchmarks

Must Practice

Class 30–40 (3), 40–50 (7), 50–60 (12), 60–70 (15), 70–80 (8), 80–90 (5)

$$\bar{x} = \Sigma fx / \Sigma f$$

$$\sigma^2 = \Sigma f(x-\bar{x})^2 / \Sigma f \quad (\text{population}) \quad s^2 = \Sigma f(x-\bar{x})^2 / (\Sigma f - 1) \quad (\text{sample})$$

Common Mistakes

1. **Population vs sample denominator** — divide by N (pop) vs n–1 (sample).
2. **Using mean for skewed data** — median is representative when skew is strong.
3. **Confusing variance with standard deviation** — always report units.
4. **Quartiles on grouped data** — use interpolation formula, not raw position.
5. **Mode of grouped data** — modal class, not the class midpoint.

Typical Questions

- 1. Compute mean, median, mode, Q1, Q3, IQR, variance, SD, C.V. for a raw dataset.
- 2. Compute mean and variance for grouped data by the midpoint method.
- 3. Which measure of central tendency is best for skewed data? Justify.
- 4. Compare dispersion of two datasets with different units using C.V.
- 5. From moments, compute skewness and kurtosis and interpret.

Examiner Expectations

Level	Performance
Pass	Computes mean/median/mode for ungrouped data
Good	Handles grouped data, variance, quartiles
Excellent	Interprets skewness/kurtosis/C.V. correctly
Full marks	All measures computed fast and error-free

Formula Sheet

Mean (raw): $\bar{x} = \Sigma x / n$
Mean (grouped): $\bar{x} = \Sigma fx / \Sigma f$
Median (raw): middle value (avg of two middles if n even)
Median (grouped): $L + (n/2 - CF) / f \times h$
Mode (grouped): $L + (f_1 - f_0) / (2f_1 - f_0 - f_2) \times h$
Variance (pop): $\sigma^2 = \Sigma (x-\mu)^2 / N = \Sigma x^2 / N - \mu^2$
Variance (sample): $s^2 = \Sigma (x-\bar{x})^2 / (n-1)$
Quartile deviation: $(Q3 - Q1) / 2$
C.V.: $(\sigma / \bar{x}) \times 100$
Moments: $\mu_r = \Sigma f (x-\bar{x})^r / \Sigma f$
Skewness (β_1): μ_3^2 / μ_2^3
Kurtosis (β_2): μ_4 / μ_2^2

GPA Priority: ■ Must Win (numerical, exam regular)

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 outlier robustness + Q3 skewness interpretation) · def/formula skim · same-problem drill #47
- **Same-problem drill target:** all descriptive measures for 2 datasets; variance pop vs sample without error
- **Deliverable:** complete descriptive statistics for 2 datasets
- **Trap:** population vs sample denominator (N vs n–1); mean is not representative for skewed data

W3 — Probability Theory · 31 Aug–06 Sep · P2 · 4 hrs

Banner: Rotation PS (Mon+Tue) · Tier P2 · Time budget 4 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 3	lines 52–68
Fear-Killer Pack	Fear-Killer-Packs.md → Week 2	lines 18–24 (Q2, Q4)
Week manifest	weeks/PS-W3.md	full file
Chapter breakdown	03-Chapter-Breakdowns/03-Probability.md	embedded below

Definitions (verbatim)

Term	Definition
Sample space (S)	The set of all possible outcomes of a random experiment.

Event	A subset of the sample space.
Mutually exclusive events	Events that cannot occur together ($A \cap B = \emptyset$).
Independent events	Events where the occurrence of one does not affect the probability of the other, $P(A \cap B) = P(A)P(B)$.
Conditional probability	$P(A B) = P(A \cap B) / P(B)$, the probability of A given that B has occurred.
Bayes' rule	$P(A_i B) = P(B A_i)P(A_i) / \sum_j P(B A_j)P(A_j)$.

Formulas (verbatim)

Formula	Statement
Permutations	$nPr = n! / (n-r)!$
Combinations	$nCr = n! / (r!(n-r)!)$
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Multiplication	$P(A \cap B) = P(A) \times P(B A)$
Conditional	$P(A B) = P(A \cap B) / P(B)$
Total law	$P(B) = \sum_i P(B A_i) P(A_i)$

Diagrams · Numericals

Type	Items
Diagrams	— (probability tree drawn from blank page; no Diagram-Book entry)
Numericals	Numerical-Book.md · #49 permutations/combinations · #50 conditional probability · #52 law of total probability

Books · Chapters · Media

Resource	Where
Devore	Ch 2 (probability)
StatQuest	Probability
Brandon Foltz	Statistics 101 — Probability playlist
Stat 110	Harvard (Blitzstein)

Fear-Killer Pack Week 2 (verbatim — Q2, Q4)

Resources: Brandon Foltz (Probability) · Stat 110 · Devore Ch.2 · William Chen Cheatsheet

- A bag contains 5 red and 7 blue marbles. Three marbles are drawn without replacement. Calculate: $P(\text{all red})$, $P(\text{exactly 2 red})$, $P(\text{at least 1 blue})$, $P(\text{first is red} \mid \text{second and third are blue})$. Solve using both combinatorics ($C(n,r)$) and sequential probability. Show that both methods agree.
- Given a contingency table of 200 people: Male/Employed=60, Male/Unemployed=20, Female/Employed=80, Female/Unemployed=40. Calculate $P(\text{Employed})$, $P(\text{Male} \mid \text{Employed})$, $P(\text{Employed} \cap \text{Male})$, $P(\text{Employed} \cup \text{Male})$. Are gender and employment status independent? Show the mathematical proof.

Narrative — Week 3 (verbatim)

Topics: Sample space; Events; Axioms of probability; Addition rule; Multiplication rule; Permutations and combinations

Resources: Devore Ch.2; StatQuest — Probability

Practice: - Solve 10 combinatorial problems (permutations, combinations) - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ - $P(A \cap B) = P(A) \times P(B|A)$ - Conditional probability from contingency tables

Deliverable: 10 probability problems (permutations, combinations, basic rules)

Time budget: 4 hrs

Chapter 3 — Probability (verbatim)

Weight: ★★☆☆☆

Difficulty: ★★☆☆☆

Importance: ★★★★★

Learning Objectives

Explain

- Basic concepts, sample space, events
- Definitions of probability (classical, relative frequency, axiomatic)

Calculate

- Permutations & combinations
- Laws of probability (addition, multiplication)

Apply

- Conditional probability
- Bayes' rule

Differentiate

- Mutually exclusive vs independent events
- Prior vs posterior probability

Must Memorize

Sample space (S): The set of all possible outcomes of a random experiment.

Event: A subset of the sample space.

Mutually exclusive events: Events that cannot occur together ($A \cap B = \emptyset$).

Independent events: Events where the occurrence of one does not affect the probability of the other, $P(A \cap B) = P(A)P(B)$.

Conditional probability: $P(A|B) = P(A \cap B) / P(B)$, the probability of A given that B has occurred.

Bayes' rule: $P(A_i|B) = P(B|A_i)P(A_i) / \sum_j P(B|A_j)P(A_j)$.

Must Understand

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; special case for mutually exclusive
- Base rate fallacy: $P(A|B) \neq P(B|A)$ — the #1 exam trap
- Permutation = ordered (nPr); combination = unordered (nCr)
- Conditional probability as renormalizing over the conditioning event
- Multiplication rule chains for sequential draws without replacement

Must Practice

Diagnostic test: sensitivity 99%, specificity 95%, prevalence 2%.

$$P(\text{disease} \mid \text{positive}) = 0.99 \times 0.02 / (0.99 \times 0.02 + 0.05 \times 0.98) \approx 0.288$$

Bag: 5 red, 7 blue. 3 draws w/o replacement.

$$P(\text{all red}) = (5/12)(4/11)(3/10)$$

$$P(\text{exactly 2 red}) = C(5,2)C(7,1)/C(12,3)$$

Common Mistakes

1. **Confusing $P(A|B)$ with $P(B|A)$** — medical-test base rate trap.
2. **Adding probabilities of non-mutually-exclusive events** — double counts intersection.
3. **Using nCr instead of nPr** — order matters for arrangements.
4. **Treating dependent events as independent** — draws without replacement.

5. Forgetting complement trick — $P(\text{at least one}) = 1 - P(\text{none})$.

Typical Questions

1. Prove Bayes' theorem from conditional probability.
2. Medical test problem with sensitivity/specificity/prevalence (base rate fallacy).
3. Draws without replacement: all red, exactly 2 red, at least 1 blue.
4. Contingency table: $P(\text{Employed})$, $P(\text{Male}|\text{Employed})$, independence test.
5. Permutations vs combinations: number of ways to arrange/select.

Examiner Expectations

Level	Performance
Pass	Computes basic and conditional probabilities
Good	Handles multiplication/addition rules and combinations
Excellent	Full Bayes' problems, proves theorems, interprets posterior
Full marks	Base-rate problems solved with probability tree, error-free

Formula Sheet

Permutations: $nPr = n! / (n-r)!$
Combinations: $nCr = n! / (r!(n-r)!)$
Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Multiplication: $P(A \cap B) = P(A) \times P(B|A)$
Conditional: $P(A|B) = P(A \cap B) / P(B)$
Total law: $P(B) = \sum_i P(B|A_i) P(A_i)$
Bayes: $P(A_i|B) = P(B|A_i) P(A_i) / \sum_j P(B|A_j) P(A_j)$

GPA Priority: ■ Must Win (Bayes guaranteed)

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q2 combinatorial + sequential + Q4 contingency) · def/formula skim · same-problem drill #49
- **Same-problem drill target:** nPr/nCr instant; $P(A \cup B)$ without double counting
- **Deliverable:** 10 probability problems (permutations, combinations, basic rules)
- **Trap:** nCr vs nPr ; dependent draws are NOT independent; complement trick $P(\text{at least one}) = 1 - P(\text{none})$

W4 — Conditional Probability & Bayes' Theorem · 07–13 Sep · P2 · 5 hrs

Banner: Rotation PS (Mon+Tue) · Tier P2 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 4	lines 70–88
Fear-Killer Pack	Fear-Killer-Packs.md → Week 2	lines 18–24 (Q1, Q3)
Week manifest	weeks/PS-W4.md	full file
Chapter breakdown	03-Chapter-Breakdowns/03-Probability.md	reference (embedded W3)

Definitions (verbatim)

Term	Definition
Conditional probability	$P(A B) = P(A \cap B) / P(B)$, the probability of A given that B has occurred.
Bayes' rule	$P(A_i B) = P(B A_i)P(A_i) / \sum_j P(B A_j)P(A_j)$.
Prior probability	Probability before new evidence (e.g., disease prevalence).

Posterior probability	Probability after conditioning on evidence (e.g., $P(\text{disease} \mid \text{positive})$).
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Formulas (verbatim)

Formula	Statement
Bayes	$P(A_i B) = P(B A_i)P(A_i) / \sum_j P(B A_j)P(A_j)$
Total law	$P(B) = \sum_i P(B A_i) P(A_i)$

Diagrams · Numericals

Type	Items
Diagrams	— (probability tree from blank page)
Numericals	<code>Numerical-Book.md</code> #51 Bayes theorem · #52 law of total probability

Books · Chapters · Media

Resource	Where
Devore	Ch 2 (Bayes)
StatQuest	Bayes' Theorem

Fear-Killer Pack Week 2 (verbatim — Q1, Q3)

Resources: Brandon Foltz (Probability) · Stat 110 · Devore Ch.2 · William Chen Cheatsheet

- A diagnostic test for a disease has 99% sensitivity ($P(\text{positive} \mid \text{disease})$) and 95% specificity ($P(\text{negative} \mid \text{no disease})$). The disease prevalence is 2%. If a person tests positive, what is the probability they actually have the disease? Show full Bayes' theorem calculation with the probability tree. This is the classic base rate fallacy — articulate why most people intuitively get this wrong.
- Prove Bayes' theorem from the definition of conditional probability. Then use the law of total probability to derive the denominator. For the medical test example above, compute $P(\text{disease} \mid \text{positive})$ if the test were perfect (100% sensitivity, 100% specificity) and if prevalence were 50% instead of 2%.

Narrative — Week 4 (verbatim)

Topics: Conditional probability; Law of total probability; Bayes' theorem; Prior vs posterior probability; Base rate fallacy

Resources: Devore Ch.2; StatQuest — Bayes' Theorem

Practice: - Bayes' theorem: $P(A|B) = P(B|A) \times P(A) / P(B)$ - Law of total probability: $P(B) = \sum P(B|A_i) \times P(A_i)$ - **Base rate fallacy:** $P(\text{disease} \mid \text{positive test})$ is NOT $P(\text{positive test} \mid \text{disease})$. This is the #1 exam trap. - Solve 10 Bayes problems (medical testing, spam detection, etc.)

Killer trap: Students confuse $P(A|B)$ with $P(B|A)$. In medical testing: $P(\text{disease}|\text{positive test})$ is typically low even when $P(\text{positive test}|\text{disease})$ is high, because the base rate of the disease is low. This is a guaranteed exam question.

Deliverable: 10 Bayes theorem problems with full step-by-step solutions

Time budget: 5 hrs

P0 floor · Drill target · Deliverable · Trap

- P0 floor:** fear-killer pack pass (Q1 medical test base rate + Q3 proof) · def/formula skim (Bayes + total law) · same-problem drill #51
- Same-problem drill target:** medical-test Bayes < 5 min; probability tree without error
- Deliverable:** 10 Bayes theorem problems with full step-by-step solutions
- Trap:** $P(A|B)$ vs $P(B|A)$ — the guaranteed exam question; base rate keeps posterior small

W5 — Random Variables & Random Processes · 14–20 Sep · P2 · 5 hrs

Banner: Rotation PS (Mon+Tue) · Tier P2 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 5	lines 90–107
Fear-Killer Pack	Fear-Killer-Packs.md → Week 3	lines 26–33 (Q1, Q2)
Week manifest	weeks/PS-W5.md	full file
Chapter breakdown	03-Chapter-Breakdowns/04-Random-Variables-and-Random-Processes.md	embedded below

Definitions (verbatim)

Term	Definition
Random variable	A function that assigns a real number to each outcome of a random experiment.
PMF (discrete)	$P(X = x_i)$ with $\sum_i P(X = x_i) = 1$.
PDF (continuous)	$f(x) \geq 0$ with $\int f(x)dx = 1$; $P(a < X < b) = \int_a^b f(x)dx$.
CDF	$F(x) = P(X \leq x)$; for continuous $F(x) = \int_{-\infty}^x f(t)dt$.
Expected value	$E[X] = \sum x_i P(x_i)$ (discrete) or $\int x \cdot f(x)dx$ (continuous).
Moment generating function	$M_X(t) = E[e^{tX}]$; moments from $M^{(k)}_X(0)$.

Formulas (verbatim)

Formula	Statement
$E[X]$ (discrete)	$\sum x_i P(X=x_i)$
$E[X]$ (continuous)	$\int x \cdot f(x) \, dx$
$\text{Var}(X)$	$E[X^2] - (E[X])^2$
M.G.F	$M_X(t) = E[e^{tX}]$, $E[X^k] = M^{(k)}_X(0)$

Diagrams · Numericals

Type	Items
Diagrams	— (PMF/PDF sketches from blank page)
Numericals	chapter 04 Must-Practice blocks (MGF verification, PDF constant k) — Numerical-Book.md resumes at #53

Books · Chapters · Media

Resource	Where
Devore	Ch 3 (random variables)
StatQuest	Random Variables
Stat 110	Harvard (Blitzstein)

Fear-Killer Pack Week 3 (verbatim — Q1, Q2)

Resources: Stat 110 • Devore Ch.3-4 • William Chen Cheatsheet

- A discrete random variable X has PMF: $P(X=1)=0.1$, $P(X=2)=0.2$, $P(X=3)=0.3$, $P(X=4)=0.25$, $P(X=5)=0.15$. Compute $E[X]$, $E[X^2]$, $\text{Var}(X)$, and the CDF at each point. Then find $P(2 \leq X < 5)$. Derive the moment generating function $M_X(t)$. Use $M_X(t)$ to find $E[X]$ and $E[X^2]$ and verify they match.
- A continuous random variable has PDF $f(x) = kx^2$ for $0 \leq x \leq 2$, 0 otherwise. Find k , compute $P(0.5 < X < 1.5)$, $E[X]$, $\text{Var}(X)$, and derive the CDF $F(x)$. Then find the median (the value m such that $P(X \leq m) = 0.5$).

Narrative — Week 5 (verbatim)

Topics: Discrete vs continuous random variables; Probability mass function (PMF); Probability density function (PDF); Cumulative distribution function (CDF); Expected value; Variance; Moment generating function (MGF)

Resources: Devore Ch.3; StatQuest — Random Variables

Practice: - Compute $E[X] = \sum x \cdot P(X=x)$ for discrete; $E[X] = \int x \cdot f(x) \, dx$ for continuous - $\text{Var}(X) = E[X^2] - (E[X])^2$ - MGF: $M_X(t) =$

$E[e^{tX}]$ - 10 problems computing expected value and variance

Deliverable: 10 random variable problems (PMF/PDF, expectation, variance)

Time budget: 5 hrs

Chapter 4 — Random Variables & Random Processes (verbatim)

Weight: ★★☆☆☆

Difficulty: ★★★★★

Importance: ★★★★★

Learning Objectives

Explain

- Discrete & continuous random variables
- Random sequences and transformations
- Introduction to random processes and time series

Calculate

- PMF, PDF, CDF, distribution functions
- Mathematical expectations, variance
- Moment generating function (M.G.F)
- Statistical averages of random processes, autocorrelation

Analyze

- Stationarity (strict & wide-sense)
- Time averaging, ergodicity
- Markov Chain and Queuing Theory

Must Memorize

Random variable: A function that assigns a real number to each outcome of a random experiment.

PMF (discrete): $P(X = x_i)$ with $\sum_i P(X = x_i) = 1$.

PDF (continuous): $f(x) \geq 0$ with $\int f(x)dx = 1$; $P(a < X < b) = \int_a^b f(x)dx$.

CDF: $F(x) = P(X \leq x)$; for continuous $F(x) = \int_{-\infty}^x f(t)dt$.

Expected value: $E[X] = \sum x_i P(x_i)$ (discrete) or $\int x f(x)dx$ (continuous).

Moment generating function: $M_X(t) = E[e^{tX}]$; moments from $M_X^{(k)}(0)$.

Stationary process: A random process whose statistical properties are invariant under time shift.

Wide-sense stationary (WSS): Constant mean and autocorrelation depending only on $\tau = t_1 - t_2$.

Ergodic process: Time averages equal ensemble averages.

Must Understand

- $\text{Var}(X) = E[X^2] - (E[X])^2$ — the computational shortcut
- M.G.F: $E[X] = M'(0)$, $E[X^2] = M''(0)$; unique M.G.F $\leftrightarrow$ unique distribution
- Strict vs wide-sense stationarity: WSS needs only mean + autocorrelation
- Autocorrelation $R_X(\tau)$ for a WSS process; $R_X(0) = E[X^2]$
- Markov property: next state depends only on current state
- Queuing: Poisson arrivals + exponential service $\rightarrow$ M/M/1 formulas

Must Practice

$P(X=1)=0.1, P(X=2)=0.2, P(X=3)=0.3, P(X=4)=0.25, P(X=5)=0.15$
 $M_X(t) = \sum e^{tx} P(x); E[X] = M'(0); E[X^2] = M''(0) \rightarrow \text{match direct sums}$

Common Mistakes

- 1. **Var(X) ≠ E[X²]** — subtract (E[X])².
- 2. **M.G.F evaluated at t = 0 directly** — differentiate first, then set t = 0.
- 3. **WSS vs strict** — WSS is weaker; strict ⇒ WSS, not conversely.
- 4. **Autocorrelation of non-stationary process** — depends on t₁ AND t₂ separately.
- 5. **Steady-state equation** — must add Σπ_i = 1; πP = π alone is underdetermined.
- 6. **Queuing arrival vs service notation** — λ arrivals, μ service; ρ = λ/μ.

Typical Questions

- 1. Compute E[X], E[X²], Var(X), CDF, M.G.F for a given PMF; verify via M.G.F.
- 2. Find k, P(0.5<X<1.5), E[X], Var(X), CDF, median for f(x) = kx² on [0,2].
- 3. Show X(t) = A cos(2πft+θ) is WSS; compute mean and autocorrelation.
- 4. Two-state Markov chain: transition diagram, 2-step probability, steady state.
- 5. M/M/1 queue: P(empty), expected queue length, probability of exactly k arrivals.

Examiner Expectations

Level	Performance
Pass	Computes E[X], Var(X), CDF for discrete/continuous
Good	Uses M.G.F, recognizes WSS
Excellent	Full random-process analysis (autocorrelation, stationarity)
Full marks	Markov/queuing and M.G.F solved under time pressure

Formula Sheet

$E[X]$ (discrete): $\sum x_i P(X=x_i)$
 $E[X]$ (continuous): $\int x f(x) dx$
 $Var(X): E[X^2] - (E[X])^2$
M.G.F: $M_X(t) = E[e^{tX}], E[X^k] = M^{(k)}(0)$
WSS: $E[X(t)] = \mu$ (const), $R_X(t_1, t_2) = R_X(t_1 - t_2)$
Autocorrelation: $R_X(\tau) = E[X(t)X(t+\tau)]$
M/M/1: $\rho = \lambda/\mu, P_0 = 1 - \rho, L = \rho/(1-\rho)$
Markov steady state: $\pi = \pi P, \sum \pi_i = 1$

GPA Priority: ■ Must Win (theory + numerical, finals regular)

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 PMF E[X]/Var/MGF + Q2 PDF k) · def/formula skim (E[X]/Var/MGF) · same-problem drill #49–52 retained for interleaving
- **Same-problem drill target:** Var(X) = E[X²] – (E[X])² < 1 min; MGF moments without error
- **Deliverable:** 10 random variable problems (PMF/PDF, expectation, variance)
- **Trap:** Var(X) ≠ E[X²]; M.G.F — differentiate first, then set t = 0

W6 — Probability Distributions · 21–27 Sep · P2 · 5 hrs

Banner: Rotation PS (Mon+Tue) · Tier P2 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.
Sleep banking begins: bedtime 22:00 for 5 nights (Weeks 6–7), sleep 9 h.

Sources & offsets

Source	Where	Ref
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Narrative	Week-by-Week-Narrative.md → Week 6	lines 108–126
Fear-Killer Pack	Fear-Killer-Packs.md → Week 4	lines 35–42 (Q1–Q5)
Week manifest	weeks/PS-W6.md	full file
Chapter breakdown	03-Chapter-Breakdowns/05-Probability-Distributions.md	embedded below

Definitions (verbatim)

Term	Definition
Binomial	$X \sim B(n, p)$ — number of successes in n independent trials, fixed p . $P(X=x) = C(n,x)p^x(1-p)^{n-x}$.
Poisson	$X \sim Po(\lambda)$ — number of rare events in a fixed interval. $P(X=x) = e^{-1}\lambda^x/x!$.
Hypergeometric	Number of successes in n draws without replacement from a finite population with K successes. $P(X=x) = C(K,x)C(N-K,n-x)/C(N,n)$.
Negative binomial	Number of trials needed to get r successes, $P(X=x) = C(x-1,r-1)p^r(1-p)^{x-r}$.
Exponential	Continuous time-between-events distribution, $f(x) = \lambda e^{-\lambda x}$, memoryless.
Normal	$X \sim N(\mu, \sigma^2)$; $Z = (X-\mu)/\sigma \sim N(0,1)$; symmetric bell curve.

Formulas (verbatim)

Formula	Statement
Binomial	$P(X=x) = C(n,x) p^x (1-p)^{n-x}$; $E = np$, $Var = np(1-p)$
Poisson	$P(X=x) = e^{-1} \lambda^x / x!$; $E = \lambda$, $Var = \lambda$
Hypergeometric	$P(X=x) = C(K,x)C(N-K,n-x)/C(N,n)$; $E = n(K/N)$
Exponential	$f(x) = \lambda e^{-\lambda x}$; $E = 1/\lambda$, $Var = 1/\lambda^2$
Normal	$Z = (X-\mu)/\sigma$; $P(Z<z)$ from table
Poisson→Binomial limit	$n \geq 20$, $p \leq 0.05$, $\lambda = np$
Normal→Binomial approx	$np \geq 5$, $n(1-p) \geq 5$

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md #1 normal distribution curve with Z-scores · #8 binomial PMF · #9 Poisson PMF
Numericals	Numerical-Book.md #53 Binomial · #54 Poisson · #55 Normal $Z = (X-\mu)/\sigma$ · #56 Z-table lookup

Books · Chapters · Media

Resource	Where
Devore	Ch 3–4 (distributions)
StatQuest	each distribution playlist
Brandon Foltz	Statistics 101 — Distributions playlist

Fear-Killer Pack Week 4 (verbatim)

Resources: StatQuest (Distributions) • Brandon Foltz (Distributions) • Devore Ch.4-5

- A factory produces chips with 2% defect rate. From a batch of 20 chips, calculate: P(exactly 1 defective) using Binomial, P(at most 2 defective), P(at least 1 defective). Now if the batch size is 500 with same defect rate, approximate using Poisson ($\lambda=10$). Compute P(exactly 5 defective) using both Binomial and Poisson and compare.
- IQ scores follow Normal(100, 15²). Calculate: P(IQ > 130), P(85 < IQ < 115), the IQ value at the 95th percentile, and P(IQ > 145 | IQ > 130). Standardize each using $Z = (X - \mu)/\sigma$ and read the Z-table.
- Explain when to use Binomial vs Poisson vs Normal. Given: (a) number of emails in an hour, (b) number of heads in 100 coin flips, (c) heights of adult women, (d) number of defective items in a batch of 10. For each, identify the correct distribution and

justify your choice.

- The lifetime of a light bulb follows Exponential($\lambda = 0.001$ failures per hour). Calculate: $P(\text{bulb lasts} > 2000 \text{ hours})$, $P(\text{bulb lasts} > 3000 \mid \text{it has already lasted } 1000)$, $E[X]$, $\text{Var}(X)$. Show that the memoryless property holds by computing $P(X > t+s \mid X > s)$ and comparing to $P(X > t)$.
- A warehouse shipment contains 50 items, of which 8 are defective. An inspector draws 10 items without replacement. Calculate $P(\text{exactly 2 defective})$ using the hypergeometric distribution. Compare this to the binomial approximation (why is it wrong here?) and to the Poisson approximation. Then: a production line has a 5% defect rate; find $P(\text{the 6th item inspected is the 3rd defective})$ using the negative binomial distribution. When must you use hypergeometric instead of binomial?

Narrative — Week 6 (verbatim)

Topics: Discrete: Binomial, Poisson, Hypergeometric, Negative Binomial; Continuous: Uniform, Exponential, Normal

Resources: Devore Ch.3-4; StatQuest — each distribution

Practice: - **When to use which:** - Binomial: n trials, success/failure, fixed p - Poisson: rare events in fixed interval ($\lambda = \text{rate}$) - Normal: symmetric, bell-shaped, mean = median = mode - Exponential: time between events (memoryless property) - Poisson approximates Binomial when $n \geq 20$ and $p \leq 0.05$

Deliverable: 10 distribution identification problems + parameter computation

Time budget: 5 hrs

Chapter 5 — Probability Distributions (verbatim)

Weight: ★★☆☆☆

Difficulty: ★★☆☆☆

Importance: ★★★★★

Learning Objectives

Explain

- Discrete vs continuous distributions
- When each distribution applies

Calculate

- Binomial, Poisson, Hypergeometric, Negative Binomial
- Uniform, Exponential, Normal (parameters, probabilities, moments)

Apply

- Normal approximation to binomial/Poisson
- Z-score standardization and Z-table reading

Differentiate

- Binomial vs hypergeometric vs Poisson vs negative binomial

Must Memorize

Binomial: $X \sim B(n, p)$ — number of successes in n independent trials, fixed p . $P(X=x) = C(n,x)p^x(1-p)^{n-x}$.

Poisson: $X \sim \text{Po}(\lambda)$ — number of rare events in a fixed interval. $P(X=x) = e^{-\lambda}\lambda^x/x!$.

Hypergeometric: Number of successes in n draws without replacement from a finite population with K successes. $P(X=x) = \frac{C(K,x)C(N-K,n-x)}{C(N,n)}$.

Negative binomial: Number of trials needed to get r successes, $P(X=x) = C(x-1,r-1)p^r(1-p)^{x-r}$.

Exponential: Continuous time-between-events distribution, $f(x) = \lambda e^{-\lambda x}$, memoryless.

Normal: $X \sim N(\mu, \sigma^2)$; $Z = (X-\mu)/\sigma \sim N(0,1)$; symmetric bell curve

Must Understand

- Binomial requires fixed n + independent trials + constant p ; without-replacement $\rightarrow$ hypergeometric
- Poisson = binomial limit when n large, p small ($n \geq 20$, $p \leq 0.05$)
- Negative binomial generalizes geometric ($r = 1$)
- Exponential is memoryless; Normal is symmetric, μ = median = mode
- Poisson approximates binomial; Normal approximates binomial when $np \geq 5$ and $n(1-p) \geq 5$

Must Practice

- (a) emails/hour $\rightarrow$ Poisson
(b) heads in 100 flips $\rightarrow$ Binomial
(c) heights of women $\rightarrow$ Normal
(d) defectives in a batch of 10 $\rightarrow$ Binomial
(e) draws without replacement from shipment $\rightarrow$ Hypergeometric

$IQ \sim N(100, 15^2)$: $P(IQ > 130) \rightarrow Z = 2 \rightarrow 0.0228$

95th percentile: $Z = 1.645 \rightarrow IQ \approx 124.7$

Common Mistakes

1. **Using binomial when sampling without replacement** — must use hypergeometric.
2. **Poisson λ mis-set** — scale λ to the interval given.
3. **Exponential memoryless misused** — $P(X > s+t | X > s) = P(X > t)$.
4. **Negative binomial confusion** — r -th success vs first success (geometric).
5. **Z-table direction** — $P(Z > z) = 1 - P(Z < z)$; tail care.
6. **Normal approx continuity correction** — skip only if asked to approximate crudely.

Typical Questions

1. Binomial: batch of 20 chips, 2% defect — $P(\text{exactly } 1)$, $P(\text{at most } 2)$, $P(\text{at least } 1)$.
2. Poisson approximation: $n = 500$, $p = 0.02$, compare $P(5 \text{ defects})$ binomial vs Poisson.
3. Hypergeometric: 50 items, 8 defective, sample 10 — $P(\text{exactly } 2 \text{ defective})$ vs binomial.
4. Negative binomial: 5% defect rate — $P(6\text{th item is the } 3\text{rd defective})$.
5. Normal: IQ percentiles, conditional probabilities.
6. Exponential: $P(X > 2000)$, conditional, $E[X]$, $\text{Var}(X)$, memoryless check.

Examiner Expectations

Level	Performance
Pass	Computes probabilities for each distribution from formula
Good	Identifies correct distribution, uses tables correctly
Excellent	Handles approximation between distributions, interprets parameters
Full marks	All distribution problems solved fast, error-free

Formula Sheet

Binomial: $P(X=x) = C(n, x) p^x (1-p)^{n-x}$; $E = np$, $\text{Var} = np(1-p)$
Poisson: $P(X=x) = e^{-\lambda} \lambda^x / x!$; $E = \lambda$, $\text{Var} = \lambda$
Hypergeometric: $P(X=x) = C(K, x) C(N-K, n-x) / C(N, n)$; $E = n(K/N)$
Negative Binom: $P(X=x) = C(x-1, r-1) p^r (1-p)^{x-r}$; $E = r/p$
Uniform: $f(x) = 1/(b-a)$; $E = (a+b)/2$
Exponential: $f(x) = \lambda e^{-\lambda x}$; $E = 1/\lambda$, $\text{Var} = 1/\lambda^2$
Normal: $Z = (X-\mu)/\sigma$; $P(Z < z)$ from table
Poisson $\rightarrow$ Binomial limit: $n \geq 20$, $p \leq 0.05$, $\lambda = np$
Normal $\rightarrow$ Binomial approx: $np \geq 5$, $n(1-p) \geq 5$

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 binomial + Poisson approx + Q2 normal) · def/formula skim (all distribution formulas) · same-problem drill #53, #54, #56
- **Same-problem drill target:** Z-table lookup < 10 s; binomial/poisson computation < 5 min; distribution identification instant
- **Deliverable:** 10 distribution identification problems + parameter computation
- **Trap:** binomial vs hypergeometric (with/without replacement); Poisson λ scaled to the interval; Z-table tail direction

W7 — Midterm Revision · 28 Sep–04 Oct · P1 · 7 hrs

Banner: Rotation PS (Mon+Tue) · Tier P1 · Time budget 7 hrs · P0 floor: pack pass + def/formula skim + same-problem drill. **No new material.**

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 7	lines 128–145
Fear-Killer Pack	Fear-Killer-Packs.md → all packs review	lines 11–42
Week manifest	weeks/PS-W7.md	full file
Chapter breakdowns	all of 01–05	reference (embedded W1, W2, W3, W5, W6)

Definitions (verbatim) — full W1–W6 sweep

Term	Definition
Statistics	The science of collecting, organizing, presenting, analyzing and interpreting data to make decisions.
Frequency distribution	A table that groups data into classes and records the number (frequency) of observations falling in each class.
Mean	The sum of all observations divided by their number.
Median	The middle value when data is arranged in order; Q2.
Standard deviation	The positive square root of the variance — the average deviation of observations from the mean.
Coefficient of variation	The ratio of standard deviation to mean, expressed as a percentage — $\sigma/\bar{x} \times 100$.
Sample space (S)	The set of all possible outcomes of a random experiment.
Conditional probability	$P(A B) = P(A \cap B) / P(B)$, the probability of A given that B has occurred.
Bayes' rule	$P(A_i B) = P(B A_i)P(A_i) / \sum_j P(B A_j)P(A_j)$.
Random variable	A function that assigns a real number to each outcome of a random experiment.
Expected value	$E[X] = \sum x_i P(x_i)$ (discrete) or $\int x f(x)dx$ (continuous).
Binomial	$X \sim B(n, p)$ — number of successes in n independent trials, fixed p. $P(X=x) = C(n,x)p^x(1-p)^{n-x}$.
Normal	$X \sim N(\mu, \sigma^2)$; $Z = (X-\mu)/\sigma \sim N(0,1)$; symmetric bell curve.

Formulas (verbatim) — full sweep

Formula	Statement
Sturges	$k = 1 + 3.322 \log_{10}(n)$
Variance (pop)	$\sigma^2 = \sum (x-\mu)^2 / N$
Variance (sample)	$s^2 = \sum (x-\bar{x})^2 / (n-1)$
Permutations	$nPr = n! / (n-r)!$
Combinations	$nCr = n! / (r!(n-r)!)$
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Bayes	$P(A_i B) = P(B A_i)P(A_i) / \sum_j P(B A_j)P(A_j)$

Var(X)	$E[X^2] - (E[X])^2$
Binomial	$P(X=x) = C(n,x) p^x (1-p)^{n-x}$
Poisson	$P(X=x) = e^{-1} \lambda^x / x!$
Normal	$Z = (X-\mu)/\sigma$

Diagrams · Numericals

Type	Items
Diagrams	Diagram-Book.md #1–#3 normal/Z curve, histogram+polygon, box plot · #8–#9 binomial/Poisson PMF
Numericals	Numerical-Book.md #46–#56 full review (descriptive → distributions)

Books · Chapters · Media

Resource	Where
Weeks 1–6 packs	timed past papers

Fear-Killer Pack — Week 7 (verbatim)

Resources: all Weeks 1-6 packs • timed past papers

- Descriptive statistics: mean, variance from raw data.
- Bayes' theorem: 3 medical testing problems.
- Distribution identification: given a scenario, name the distribution and parameters.
- Z-score: $P(Z < z)$, $P(Z > z)$, $P(z_1 < Z < z_2)$ from standard normal table.

Narrative — Week 7 (verbatim)

Topics: Comprehensive review of Weeks 1-6

Practice: - Timed past paper (2 hrs, closed book) - Descriptive statistics: mean, variance from raw data - Bayes' theorem: 3 medical testing problems - Distribution identification: given a scenario, name the distribution and parameters - Z-score: $P(Z < z)$, $P(Z > z)$, $P(z_1 < Z < z_2)$ from standard normal table

Red flag: If Z-table reading takes >30 seconds per lookup, practice until it takes 10 seconds. The table is your best friend.

Sleep banking: Bedtime moves to 22:00 for 5 nights (Weeks 6-7). Sleep 9 hours. Formula-dense content benefits especially from NREM slow-wave consolidation — protect your Bayes and distribution memory.

Time budget: 7 hrs

P0 floor · Drill target · Deliverable · Trap

- P0 floor:** fear-killer pack pass (Q1 descriptive + Q2 Bayes + Q4 Z-table) · def/formula skim (full W1–W6 sweep) · same-problem drill #51/#56
- Same-problem drill target:** Z-lookup < 10 s; Bayes 3 problems < 15 min; distribution identification instant
- Deliverable:** timed past paper (2 hrs, closed book) solved and self-graded
- Trap:** Z-table fluency is the gate — 10 lookups in 2 minutes at W8; don't re-read notes, retrieve

W8 — MIDTERM EXAM WEEK · 05–11 Oct · Exam

Banner: Rotation PS (Mon+Tue) · Tier MIDTERM · No new deep study, no floor accrual. Ledger frozen during W8.

Sources

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 8	lines 147–156
Pack	Fear-Killer-Packs.md → Week 8	no pack
Week manifest	weeks/PS-W8.md	full file

Fear-Killer Pack — Week 8 (verbatim)

Narrative — Week 8 (verbatim)

Focus: Active recall only. No new material.

- Review formula sheet (prepared Week 7)
- Z-table speed drills: 10 lookups in 2 minutes
- Bayes: 2 complete problems
- Sleep 8 hours each night

Exam-day stack (Mon–Fri)

- [] Past-paper run for the exam subject — 60 min, P0
- [] Blank-page retrieval of that subject — 30 min, P0
- [] Master Error Log review — 20 min, P0

Notes

- Review the 1-page formula sheet (prepared Wk7). Sleep 8 h each night.
- Exam order: numericals first, theory second.
- **Ledger frozen during Wk8** — no new accrual; cleared in Wk9 recovery.